

RASPBERRY PI SETUP GUIDE (ZERO 2 W)

Computing with Words

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Chapter 1

Raspberry Pi Setup and SSH Connection

1.1 Overview

This document describes the recommended first-time setup workflow for a headless Raspberry Pi Zero 2 W used in the course. The goal is to prepare the microSD card, boot the Pi, connect over Wi-Fi, and verify that the environment is ready for GPIO-based exercises.

1.2 What is a Raspberry Pi and why we use it?

A Raspberry Pi is a small computer that runs Linux (Raspberry Pi OS). In this course, you will use it to connect and control real hardware (LEDs, sensors, speakers) and to run small programs that read inputs and produce outputs.



Figure 1.1: Raspberry Pi Zero 2 W.

1.3 Raspberry Pi Zero 2 W: main parts

- **microSD slot:** Where the operating system lives. The Pi boots from the microSD card.
- **micro-USB PWR IN:** Power input. Use this port to power the Pi.

- **micro-USB USB/OTG:** Data port for USB devices (keyboard, adapters) using an OTG cable.
- **mini-HDMI:** Video output to a monitor.
- **40-pin GPIO header:** Pins used to connect sensors/actuators (input/output).

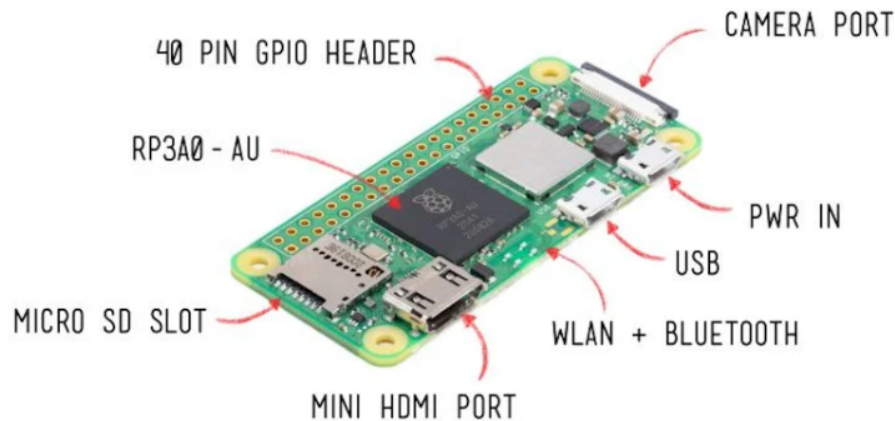


Figure 1.2: Raspberry Pi Zero 2 W.

1.4 Key terms

- **SSH:** A secure way to control the Pi from your terminal over Wi-Fi (no monitor needed).
- **Hostname:** A name for the Pi on the network (example: `cwpi-zero2w`). Often used as `cwpi-zero2w.local`.
- **GND:** Ground (0V). Many circuits must connect to GND to work.
- **3V3 / 5V:** Power pins. GPIO *signal* pins use 3.3V logic (do not feed 5V into GPIO signal pins).
- **GPIO:** Programmable pins used to read sensors (input) or control devices like LEDs (output).
- **BCM numbering:** Pin naming scheme such as `GPIO17`. This guide uses BCM numbers.

1.5 Goal

- Install Raspberry Pi OS onto a microSD card using Raspberry Pi Imager.
- Boot the Raspberry Pi Zero 2 W from the microSD card.
- Connect from macOS to the Pi using SSH over Wi-Fi.
- Run one short Python command to confirm GPIO tools are working.

1.6 Requirements

1.6.1 Hardware

- Raspberry Pi Zero 2 W
- microSD card (**32 GB recommended**)
- microSD card reader/adapter for your computer
- micro-USB power cable + power adapter (use **PWR IN** on the Pi)
- Phone hotspot (**recommended** if dorm/uni Wi-Fi uses a login page)

1.6.2 Software

- **Raspberry Pi Imager** (official)

1.6.3 Network recommendation (hotspot vs campus Wi-Fi vs LAN)

University or student dorm Wi-Fi networks often require a browser login page (captive portal). A headless Raspberry Pi cannot easily complete that login during first boot. A phone hotspot is usually the simplest reliable option for the first-time setup. If compatible and available, you can also use a **LAN/Ethernet connection** (via an adapter) for a more stable first setup.

Chapter 2

Write Raspberry Pi OS using Raspberry Pi Imager

2.1 Raspberry Pi Imager setup

2.1.1 Turn on your phone hotspot

Turn ON hotspot on your phone, note SSID (name) + password, and connect your laptop to the same hotspot.

2.1.2 Install and open Raspberry Pi Imager

Insert the microSD into the reader/adapter, then plug it into your Mac.

2.1.3 Raspberry Pi Imager

Choose Device

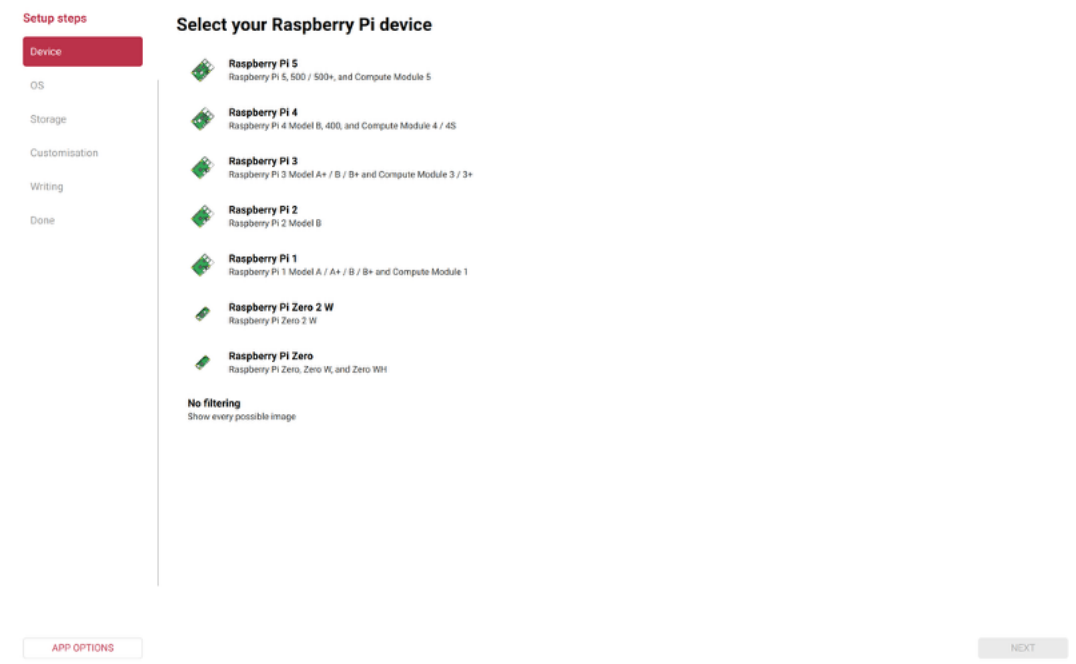


Figure 2.1: Choose Device → select Raspberry Pi Zero 2 W.

Choose OS

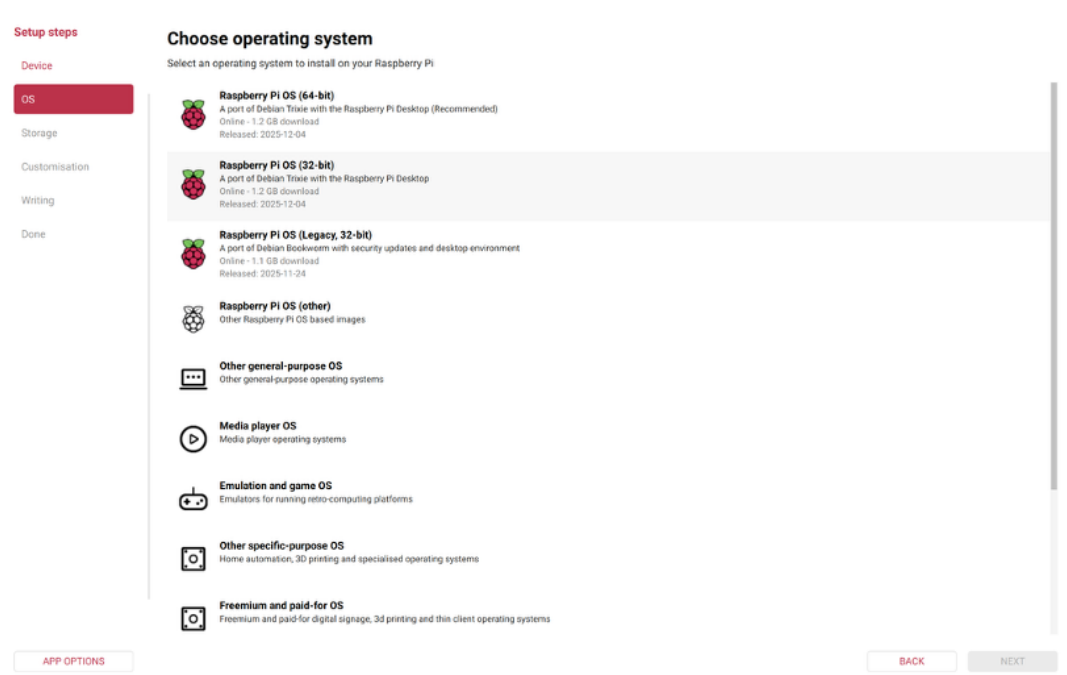


Figure 2.2: Choose OS → select Raspberry Pi OS (64-bit)

Choose Storage (microSD card)

Select your microSD card as the storage device.



Figure 2.3: Choose Storage → select the microSD card

Customisation (Hostname, User, Wi-Fi, SSH)

Fill these values (recommended):

- Hostname: cwpi-zero2w
- Username: pi (or your choice)
- Wi-Fi: your phone hotspot SSID (name) + password (Country: CH)
- Remote access: Enable SSH (password authentication)

The screenshot shows the 'Customisation: Choose hostname' screen. On the left, a 'Setup steps' sidebar lists: Device, OS, Storage, Customisation (highlighted), Hostname, Localisation, User, Wi-Fi, Remote access, Raspberry Pi Connect, Writing, and Done. The main area has the title 'Customisation: Choose hostname' and a text input field with the placeholder 'Enter your hostname' and the value 'cwpi-zero2w'. Below the input field, a note states: 'A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.' At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

Figure 2.4: Customisation → Hostname (enter: cwpi-zero2w).

The screenshot shows the 'Customisation: Localisation' screen. The 'Setup steps' sidebar on the left is identical to the previous screen, with 'Localisation' highlighted. The main area has the title 'Customisation: Localisation' and the instruction 'Select your location for suggested time zone and keyboard layout'. It features three dropdown menus: 'Capital city' (selected: Bern (Switzerland)), 'Time zone' (selected: Europe/Zurich), and 'Keyboard layout' (selected: us). At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

Figure 2.5: Customisation → Location Selection.

Setup steps

Device
OS
Storage
Customisation
Hostname
Localization
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Done

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Enter your username

pi

Password:

Secret (hidden) → have look to help

.....

Confirm password:

Re-enter to change password

.....

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Figure 2.6: Create user account - remember the entered credential

Setup steps

Device
OS
Storage
Customisation
Hostname
Localization
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Done

Customisation: Choose Wi-Fi

SECURE NETWORK

OPEN NETWORK

SSID:

Network name

Password:

Network password

Confirm password:

Re-enter password

Enter a password

☐ Hidden SSID

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Figure 2.7: Enter wi-fi details SSID (name) and password (keep the credentials exactly same as wi-fi)

Setup steps

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: SSH authentication

Configure SSH access

Enable SSH

Learn about SSH

Authentication mechanism:

☒ Use password authentication

☐ Use public key authentication

Enable SSH

Learn about SSH

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Connect (optional)

Can Leave it OFF

Setup steps

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Raspberry Pi Connect

Sign in to receive a token and enable Raspberry Pi Connect

Enable Raspberry Pi Connect

What is Raspberry Pi Connect?

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Write the image

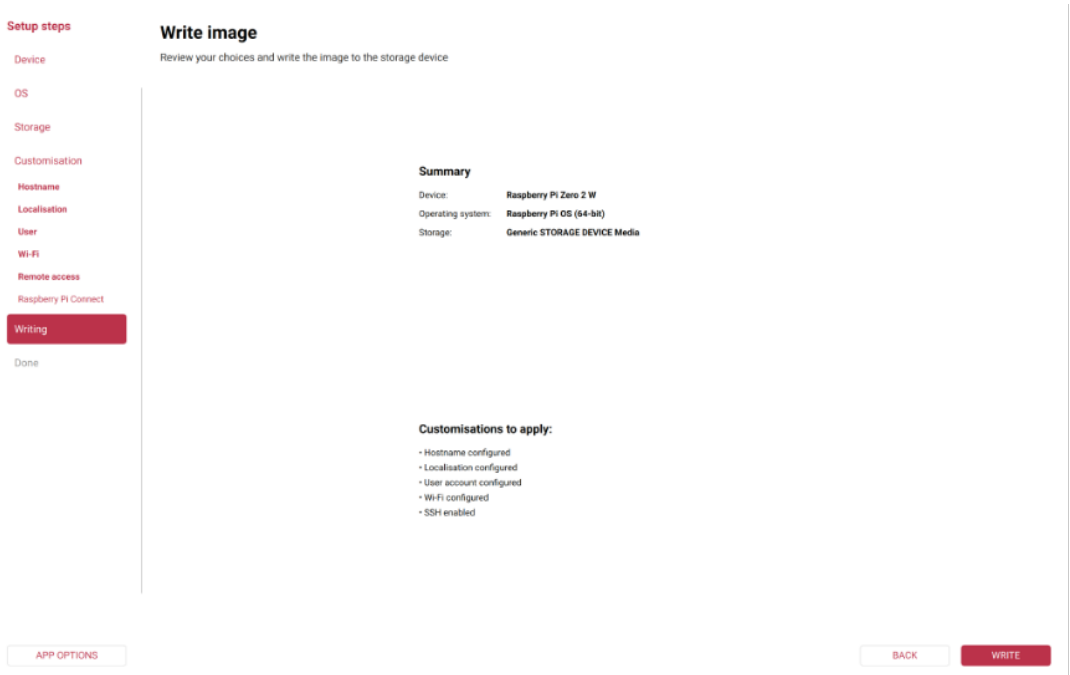


Figure 2.8: Write image → final review before writing.

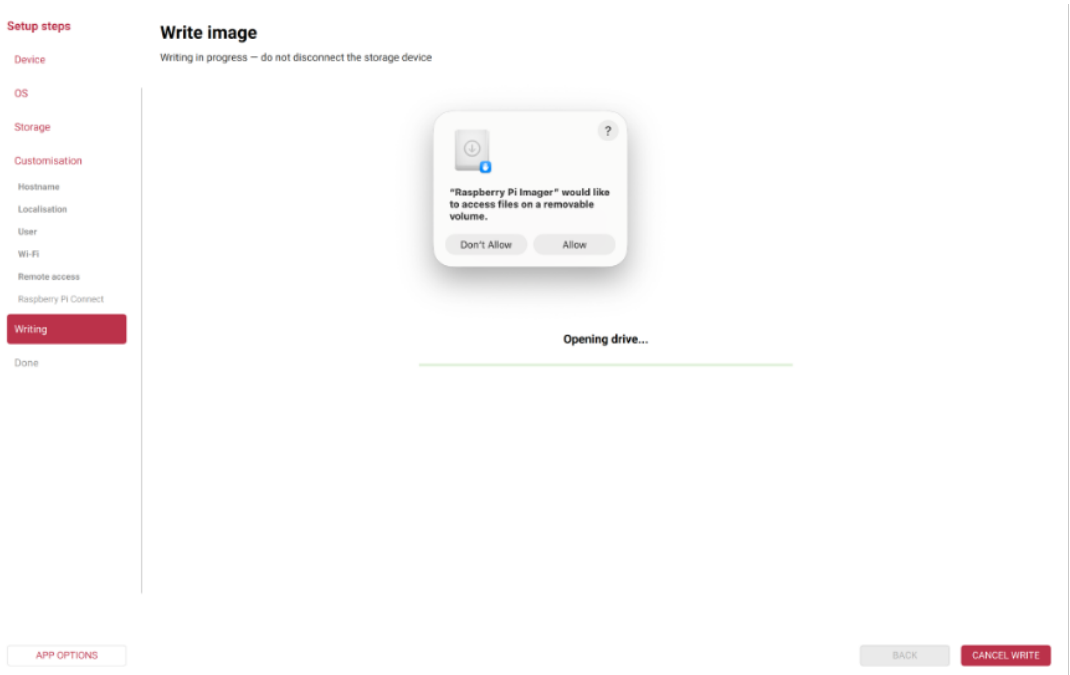


Figure 2.9: permission popups → click Allow.



Figure 2.10: Write complete → remove the card safely.

Note:

- **macOS:**

If macOS shows “The disk you attached was not readable...”, click “Ignore” (do not “Initialize”). This is normal after writing Raspberry Pi OS.

If the write process is interrupted or Raspberry Pi Imager shows an error, remove the SD card, re-insert it firmly, and run Raspberry Pi Imager again from the start. In most cases, Raspberry Pi Imager will remember your previously entered customisation settings (hostname, Wi-Fi, SSH, user), so you can re-apply them quickly.

- **Windows:**

After writing the SD card, Windows may pop up a message like “You need to format the disk in drive X: before you can use it.” Click “Cancel” (do not “Format”). This is normal because Windows cannot read the Linux partitions on the SD card.

If the write process is interrupted or Raspberry Pi Imager shows an error, remove the SD card, re-insert it firmly, and run Raspberry Pi Imager again from the start. Raspberry Pi Imager often remembers your previously entered customisation settings, so you can re-apply them quickly.

- **Linux:**

After writing the SD card, your Linux system may show a notification such as “Unknown/un-supported filesystem” or offer to mount the new partitions. You can safely “dismiss/ignore”, this is normal because the SD card now contains Linux partitions intended for the Raspberry Pi.

If the write process is interrupted or Raspberry Pi Imager shows an error, remove the SD card, re-insert it firmly, and run Raspberry Pi Imager again from the start. Raspberry Pi Imager often remembers your previously entered customisation settings, so you can re-apply them quickly.

2.1.4 Boot the Raspberry Pi

Power on

1. Insert the **microSD card** into the Raspberry Pi Zero 2 W.
2. Plug the micro-USB power cable into the **PWR IN** port.

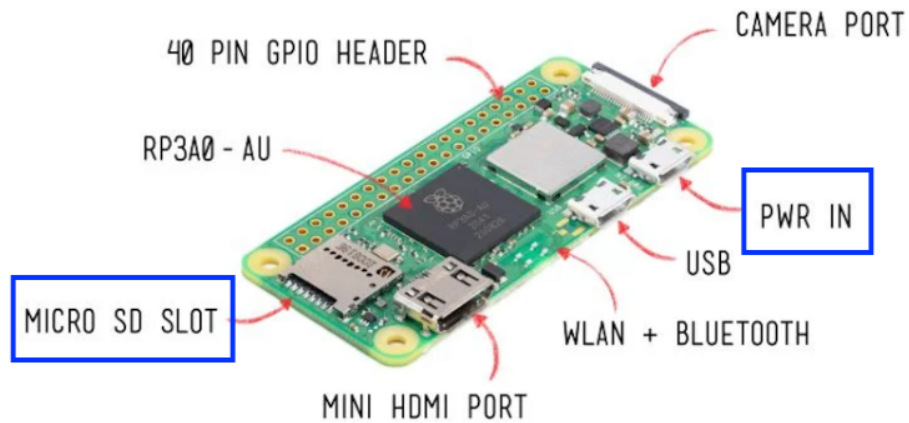


Figure 2.11: Write complete → remove the card safely.

3. Wait **2–3 minutes** for the first boot.

Chapter 3

Connect to the Raspberry Pi

3.1 Connect using SSH

3.1.1 Before you start

- Ensure your Raspberry Pi and your laptop are on the **same network** (typically your phone hotspot).
- Local name often works as: `cwpi-zero2w.local`

3.1.2 macOS (Terminal)

1. Open **Terminal**.
2. Run:

```
ssh pi@cwpi-zero2w.local
```
3. Type **yes** if asked, then enter your password.
4. If `.local` does not work, use the IP address shown in your hotspot connected devices list:

```
ssh pi<IP_ADDRESS>
```

3.1.3 Windows (PowerShell or PuTTY)

Option A: PowerShell (recommended)

1. Open **Windows PowerShell**.
2. Run:

```
ssh pi@cwpi-zero2w.local
```
3. Type **yes** if asked, then enter your password.
4. If `.local` does not work, use the IP address from your hotspot connected devices list:

```
ssh pi<IP_ADDRESS>
```

Option B: PuTTY

1. Open **PuTTY**.
2. Fill:
 - **Host Name:** `cwpi-zero2w.local` (or `<IP_ADDRESS>`)
 - **Port:** 22
 - **Connection type:** SSH
3. Click **Open**, then login as `pi` and enter your password.

3.1.4 Linux (Terminal)

1. Open a terminal.
2. Run:

```
ssh pi@cwpi-zero2w.local
```
3. Type `yes` if asked, then enter your password.
4. If `.local` does not work, use the IP address shown in your hotspot connected devices list:

```
ssh pi<IP_ADDRESS>
```

Update and Test GPIO Tools

Note: These commands are run **inside the Raspberry Pi** (after you have connected via SSH).

After SSH login (you will be able to see a prompt like: `pi@cwpi-zero2w:~ $`), run both the commands below:

```
pi@cwpi-zero2w:~ $ sudo apt-get update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Hit:4 http://archive.raspberrypi.com/debian trixie InRelease
Reading package lists... Done
```

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4.1.2 Install gpiozero (often already installed)

Run:

```
sudo apt-get install -y python3-gpiozero
```

4.1.3 Quick test

Run:

```
python3 -c "import gpiozero; print('gpiozero ok')"
```

```
pi@cwpi-zero2w:~ $ sudo apt-get install -y python3-gpiozero
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-gpiozero is already the newest version (2.0.1-0+rpt1+trixie).
0 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
pi@cwpi-zero2w:~ $ python3 -c "import gpiozero; print('gpiozero ok')"
```

Expected output:

```
gpiozero ok
```

4.2 Safe shutdown

Do not unplug power directly. Run on the Raspberry Pi:

```
sudo shutdown now
```

4.3 Troubleshooting checklist

- SSH fails: confirm Pi and laptop are on the **same Wi-Fi/hotspot**.
- cwpi-zero2w.local fails: use `ssh pi@<IP_ADDRESS>`.
- First boot is slow: wait another 1–2 minutes and retry.
- Updates fail: confirm the hotspot has internet access.